



## **REPORT**

# **“Creating an Encyclopedia for IPCC AR6 WG2 Chapter 7”**

Submitted by

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# 1. Introduction

Climate change is one of the most pressing global challenges of the 21st century, impacting ecosystems, human health, economies, and societies worldwide. Understanding its causes, impacts, and mitigation strategies requires the systematic collection, analysis, and synthesis of vast amounts of scientific data. The **Intergovernmental Panel on Climate Change (IPCC)**, established in 1988 by the United Nations, serves as the leading international body for assessing the science related to climate change.

The **IPCC Sixth Assessment Report (AR6)**, released in stages between 2021 and 2022, provides the most comprehensive and up-to-date scientific assessment of the climate system, the observed and projected changes, and the risks associated with global warming. AR6 combines findings from thousands of peer-reviewed studies, covering topics such as temperature trends, extreme weather events, sea-level rise, and the impacts on human and natural systems. Its structured chapters and extensive datasets make it an invaluable resource for policymakers, researchers, and climate advocates.

By converting the complex scientific content of AR6 into structured semantic forms and developing a corresponding dictionary or encyclopedia, it becomes possible to make climate data more **accessible, searchable, and understandable** to a wider audience, enabling informed decision-making and effective climate action.

## #semanticClimate

Semantic climate is a global platform led by climate enthusiasts with the aim to convert the complex climate data into semantic form to make it understandable and accessible to anyone around the world. Currently, the focus is on the UN's IPCC reports, particularly the sixth assessment report (AR6), which contains around 10,000 pages that need to be simplified for better understanding.

So, for this #semanticClimate has developed toolkit which include *amilib*, *pygetpapers* and *docanalysis*.

- a. ***amilib*** : It is a Python library designed for document processing, dictionary creation, and analysis of scholarly content. It provides tools for finding, cleaning, converting, searching, and republishing legacy documents in formats like PDF, PNG, HTML, and XML.
- b. ***pygetpapers***: It is a command-line tool designed to assist text miners and researchers by systematically retrieving open-access scientific literature. It connects to open access

repositories, processes search queries, analyzes the results, and downloads research papers and their metadata in bulk with minimal manual effort.

- c. **docanalysis** : A command-line tool for processing and analyzing document collections (CProjects). It supports text mining and NLP to extract meaningful information from large datasets.

With a background in healthcare, I chose to focus on **IPCC Working Group II Chapter 7: Health, Wellbeing and the Changing Structure of Communities**, to better understand how climate change affects global health, a topic often overlooked.

## 2. Objectives

- a) **Keyphrase Extraction and Analysis:**  
To extract and analyze keyphrases from the assigned chapter for efficient data organization and semantic representation.
- b) **Dictionary Development:**  
To develop a structured and comprehensive dictionary using the generated wordlist for improved terminology management and standardization.
- c) **Alpha Testing of #semanticClimate Tools:**  
To perform alpha testing of the #semanticClimate tools on operating system – macOS, identify potential issues, and report errors for debugging and performance optimization.

## 3. Detailed Overview of Work: Chapter 7 – Health, Wellbeing, and the Changing Structure of Communities

### I. Summary

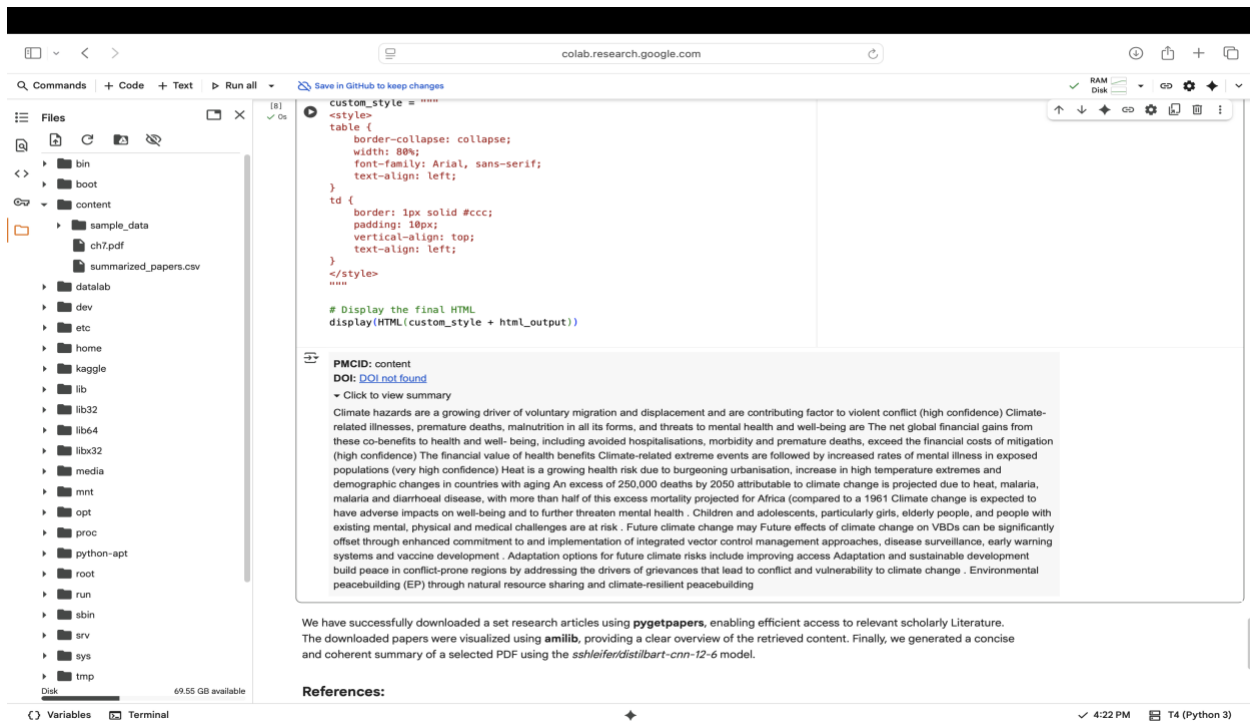
Climate hazards are increasingly driving voluntary migration and displacement and are also a contributing factor to violent conflict (high confidence). Climate change negatively affects human health by increasing the risk of illnesses, premature deaths, malnutrition, and mental health challenges. Extreme events are often followed by higher rates of mental illness in affected populations (very high confidence). Heat-related health risks are rising due to urbanization, more frequent extreme temperatures, and aging populations. By 2050, climate change is projected to cause an excess of 250,000 deaths annually, largely from heat stress, malaria, and diarrheal diseases, with over half of these deaths expected to occur in Africa (compared to 1961 levels). Children, adolescents (especially girls), elderly individuals, and people with existing physical or mental health conditions are particularly vulnerable. However, future impacts on vector-borne diseases can be significantly reduced through integrated vector control, disease surveillance,

early warning systems, and vaccine development. Adaptation and sustainable development are key to reducing conflict and building peace in climate-vulnerable regions. Environmental peacebuilding through natural resource sharing and climate-resilient practices can address the root causes of grievances and reduce vulnerability to climate change. Importantly, the net global financial benefits of health-related co-benefits, such as fewer hospitalizations and deaths, are expected to exceed the costs of climate change mitigation (high confidence).

This summary was generated in Google Colab and subsequently validated for accuracy.

*Colab link for PDF summarization:*

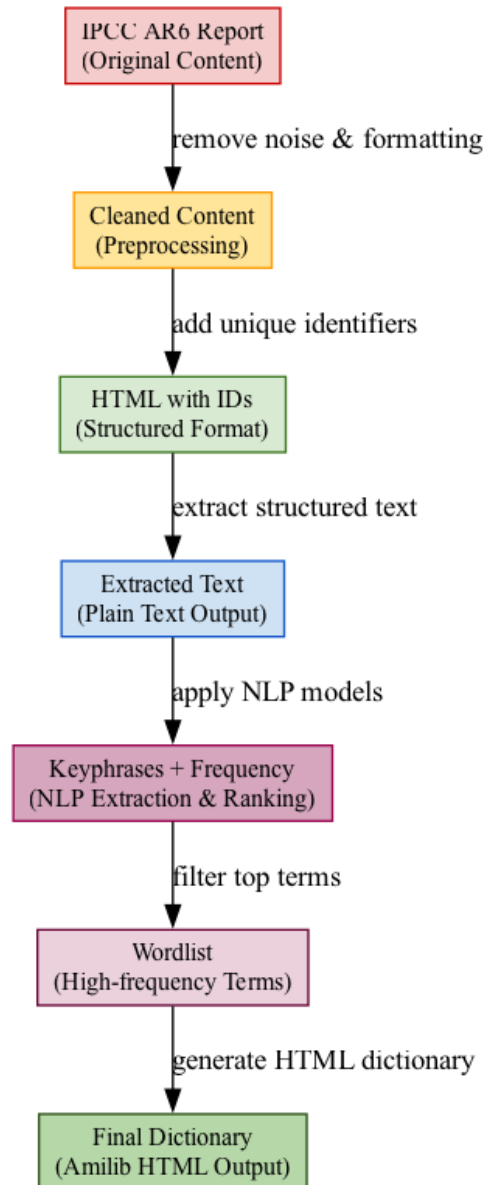
[https://colab.research.google.com/github/semanticClimate/PDF\\_Summarization\\_demo/blob/main/RevLit\\_PDF\\_Summarization.ipynb](https://colab.research.google.com/github/semanticClimate/PDF_Summarization_demo/blob/main/RevLit_PDF_Summarization.ipynb)



*Fig 1 Google Colab notebook used for chapter summarization.*

## *Workflow for Dictionary Creation Using Graphviz*

Keyword Extraction and Dictionary Creation Workflow



## II. Keyphrases extraction

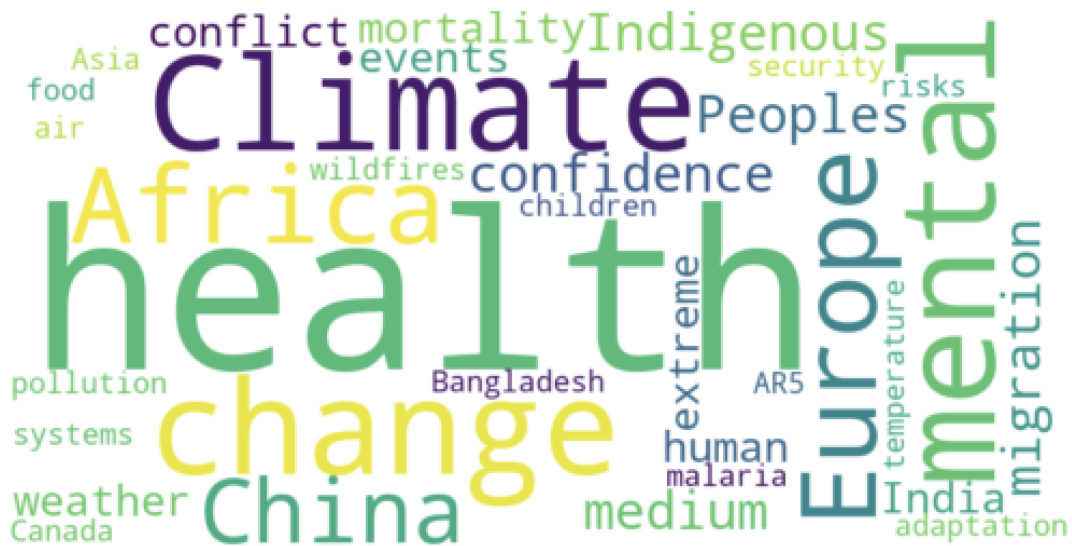
Another key responsibility was the extraction of keyphrases from the chapter text. Extracting keyphrases helps identify the most important topics and concepts within the chapter. For this, I utilized a Python-based keyword extraction tool that leverages state-of-the-art Natural Language Processing (NLP) models. This tool uses Hugging Face transformer model to automatically identify and extract the most relevant keywords, followed by frequency analysis to determine their occurrence and significance within the text.

*Link for wordlist generated*

[https://github.com/semanticClimate/internship\\_sC/blob/Shaik-Zainab/IPCC\\_WG2\\_Chapter07/ch7\\_wordlist.txt](https://github.com/semanticClimate/internship_sC/blob/Shaik-Zainab/IPCC_WG2_Chapter07/ch7_wordlist.txt)

|    |                    |
|----|--------------------|
| 1  | Climate change     |
| 2  | mental health      |
| 3  | Africa             |
| 4  | Europe             |
| 5  | WHO                |
| 6  | China              |
| 7  | Indigenous Peoples |
| 8  | health             |
| 9  | migration          |
| 10 | medium confidence  |

*Fig 2 Wordlist generated from the extracted keyphrases.*



*Fig 2.1 Visual representation of key terms using a wordcloud based on the generated wordlist.*

### **III. Dictionary creation**

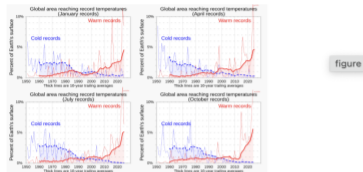
After generating keyphrases with their frequencies, I selected the high-frequency terms to create a focused wordlist for dictionary generation. Using Amilib, I produced an HTML dictionary that included Wikipedia descriptions and relevant figures for each keyword. The dictionary serves as a centralized repository of all extracted keywords, allowing quick identification of significant terms. Each keyword is linked to definitions and figures from Wikipedia and provided in multiple formats, ensuring consistent understanding and supporting further research.

*Link for Dictionary generated:*

[https://github.com/semanticClimate/internship\\_sC/blob/Shaik-Zainab/IPCC\\_WG2\\_Chapter07/ch7\\_dict.html](https://github.com/semanticClimate/internship_sC/blob/Shaik-Zainab/IPCC_WG2_Chapter07/ch7_dict.html)

search term: extreme temperatures [Wikipedia Page](#)

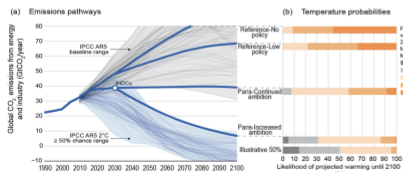
The **list of weather records** includes the most extreme occurrences of [weather phenomena](#) for various categories. Many weather records are measured under specific conditions—such as surface temperature and wind speed—to keep consistency among measurements around the Earth. Each of these records is understood to be the record value officially observed, as these records may have been exceeded before modern [weather instrumentation](#) was invented, or in remote areas without an official [weather station](#). This list does not include [remotely sensed observations](#) such as satellite measurements, since those values are not considered official records.<sup>[2]</sup>



In recent decades, new high temperature records have substantially outpaced new low temperature records on a growing portion of Earth's surface.<sup>[1]</sup> Comparison shows seasonal variability for record increases.

search term: climate change scenarios [Wikipedia Page](#)

A **climate change scenario** is a hypothetical future based on a "set of key driving forces".<sup>[1]</sup>:1812 Scenarios explore the long-term effectiveness of [mitigation](#) and [adaptation](#).<sup>[2]</sup> Scenarios help to understand what the future may hold. They can show which decisions will have the most meaningful effects on [mitigation](#) and [adaptation](#).



Four climate change scenarios, based on 2015 data.<sup>[5][6]</sup> Left: emissions pathways following the scenarios of (1) no policy, (2) current policy, (3) meeting the governments' announcements with constant country decarbonization rates past 2030, and (4) meeting the governments' announcements with higher rates of decarbonization past 2030. Right: global temperatures, depending on the amount of [greenhouse gases emitted](#) in each of the four scenarios.

search term: floods [Wikipedia Page](#)

A **flood** is an overflow of water (or rarely other fluids) that submerges land that is usually dry.<sup>[1]</sup> In the sense of "flowing water", the word may also be applied to the inflow of the [tide](#). Floods are of significant concern in [agriculture](#), [civil engineering](#) and [public health](#). [Human changes to the environment](#) often increase the intensity and frequency of flooding. Examples for human changes are [land use changes](#) such as [deforestation](#) and [removal of wetlands](#), changes in waterway course or [flood controls](#) such as with [levees](#). Global environmental issues also influence causes of floods, namely [climate change](#) which causes an [intensification of the water cycle](#) and [sea level rise](#).<sup>[2]:1517</sup> For example, climate change makes [extreme weather events](#) more frequent and stronger.<sup>[3]</sup> This leads to more intense floods and increased flood risk.<sup>[4][5]</sup>



Urban flooding in a street in Morpeth, England, 2008

search term: infectious diseases [Wikipedia Page](#)

An **infection** is the invasion of [tissues](#) by [pathogens](#), their multiplication, and the reaction of [host](#) tissues to the infectious agent and the [toxins](#) they produce.<sup>[1]</sup> An **infectious disease**, also known as a **transmissible disease** or **communicable disease**, is an [illness](#) resulting from an infection.

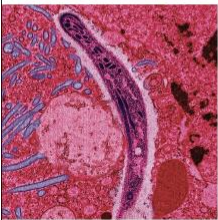


Fig 3 Dictionary generated in HTML format, displaying structured terms and associated visuals.



One of the key tasks was alpha testing the toolkit on operating system - macOS, with all identified errors documented and reported to support debugging and optimize system performance, ensuring the toolkit's functionality, stability, and compatibility across the target environment prior to broader use.

*Fig 4.1 Command-line output showing the test summary for the amilib using pytest*

*Fig 4.2 Command-line execution of pygetpapers for Chapter 7*

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                       |                                                             |                                                                                                                                                                           |                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                           |
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| <div><div><div>eupmc_result.html</div><div>eupmc_results.json</div><div>europa_pmc.csv</div></div><div><div>PMCID419630</div><div>PMCID9973115</div><div>PMCID11489660</div><div>PMCID11504743</div><div>PMCID11742948</div><div>PMCID11882091</div><div>PMCID11883379</div><div>PMCID11954265</div><div>PMCID11954788</div><div>PMCID12041194</div><div>PMCID12317649</div><div>PMCID12153470</div><div>PMCID12156489</div><div>PMCID12156562</div><div>PMCID12156700</div><div>PMCID12156733</div><div>PMCID12174194</div><div>PMCID12178532</div><div>PMCID12222477</div><div>PMCID12244781</div></div></div> | <div><div>PMCID11489660</div><div>39424906</div><div>MED</div><div>39424906</div><div>PMCID11489660</div><div>{fullTextId: PMCID11489660}</div></div> | <div><div>10.1038/s41598-024-73872-0</div><div></div></div> | <div><div>Community structure of trophic emerging from spatio-temporal variations in the Inter-tropical Convergence Zone dynamics.</div><div></div></div>                 | <div><div>Chopra G, Umri VR, Venkatesan P, Vallejo-Bernal SM, Marwan N, Kurths J, Sujith RI</div><div></div></div> | <div><div>SM, firstName: Sara M, lastName: Vallejo-Bernal, initials: SM, authorAffiliationDetailsList: {authorAffiliation: {affiliation: Potsdam Institute for Climate Impact Research (PIK), Member of the Leibniz Association, Potsdam, 14412, Germany}, affiliation: Institute of Geosciences, University of Potsdam, Potsdam, 14476, Germany}}}, {fullName: Telangana State Agricultural University, Hyderabad, Telangana, India}}, {fullName: Siddiqua A, firstName: Asma, lastName: Siddiqua, initials: A, authorAffiliationDetailsList: {authorAffiliation: {affiliation: ICAR-Central Research Institute for Dryland Agriculture, Hyderabad, Telangana, India}}, firstName: Sharma KL, lastName: K. L., bioName: Sharma, initials: KL, authorAffiliationDetailsList:</div><div>{authorId: ORCID, #text: 0003-481138X}}</div></div> |                           |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <div><div>PMCID11742948</div><div>39834453</div><div>MED</div><div>39834453</div><div>PMCID11742948</div><div>{fullTextId: PMCID11742948}</div></div> | <div><div>10.3389/fnut.2024.1444962</div><div></div></div>  | <div><div>Change in mineral composition and cooking quality in legumes grown on semi-arid alluvial due to elevated CO<sub>2</sub> and temperature.</div><div></div></div> | <div><div>Shankar KS, Vanaja M, Shankar M, Siddiqua A, Sharma KL, Giritjaveni V, Singh VK</div><div></div></div>   | <div><div>Telangana State Agricultural University, Hyderabad, Telangana, India}}, {fullName: Siddiqua A, firstName: Asma, lastName: Siddiqua, initials: A, authorAffiliationDetailsList: {authorAffiliation: {affiliation: ICAR-Central Research Institute for Dryland Agriculture, Hyderabad, Telangana, India}}, firstName: Sharma KL, lastName: K. L., bioName: Sharma, initials: KL, authorAffiliationDetailsList:</div><div>{authorId: ORCID, #text: 0003-481138X}}</div></div>                                                                                                                                                                                                                                                                                                                                                       | <div><div>NaN</div></div> |

**Fig 4.2.a** Output obtained from the `pygetpapers` query

## ➤ Tools and technologies used:

- **Python:** A powerful high-level programming language that is ideal for scripting and rapid application development.
- **VS Code:** IDE for coding, testing, and debugging.
- **Git & GitHub:** Version control and collaborative code hosting.
- **amilib:** Tool for text extraction and dictionary generation.
- **pygetpapers:** Text mining tool for systematically fetching scientific articles.
- **Google Colab:** Hosted Jupyter Notebook environment for prototyping and experiments.
- **Graphviz:** Visualization tool for networks and dependencies.
- **Slack:** Team communication and coordination platform.

## Conclusion

During the first four weeks of internship with #semanticClimate, I developed skills beyond my core healthcare background, learning the basics of Python, VS Code, Git, Google Colab, and text mining tools like amilib and pygetpapers. Collaborating with team members on debugging and development sessions enriched my understanding of these tools and their applications. Through tasks such as keyphrase extraction, dictionary creation, and alpha testing, I gained practical experience in NLP, data processing, and version control. This internship enhanced both technical and problem-solving skills, broadened my knowledge of climate data analysis, and prepared me to apply these tools effectively in future research and projects.

## Acknowledgement

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## References

<https://github.com/semanticClimate>

<https://github.com/petermr/amilib>

<https://github.com/petermr/pygetpapers>